

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEOE505A - EMBEDDED SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 An embedded system using an ARM processor needs to reduce memory access latency. Which architectural feature should be preferred?
 - A. Harvard architecture with separate instruction and data buses
 - B. Von Neumann architecture with single bus
 - C. External memory interfacing only
 - D. Serial bus architecture
- 2 In a System-on-Chip (SoC) design for an alarm clock, integrating peripherals on-chip primarily helps to:
 - A. Increase clock frequency
 - B. Reduce power consumption and board size
 - C. Increase instruction cycle count
 - D. Eliminate the need for firmware
- 3 If multiple peripherals require different bandwidths in a scalable bus architecture, which bus feature is most suitable?
 - A. Fixed-priority arbitration
 - B. Time-triggered scheduling
 - C. Hierarchical bus structure
 - D. Single master configuration
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- 5 To control the direction of rotation of a DC motor in an embedded system, which circuit is typically used?
 - A. Servo controller
 - B. Stepper driver
 - C. H-Bridge
 - D. Timer counter

- 6 In an RTOS-based embedded system, which task state allows immediate execution when CPU becomes free?
 - A. Blocked
 - B. Suspended
 - C. Ready
 - D. Dormant
- 7 A real-time task misses its deadline due to excessive interrupt latency. Which RTOS feature can help reduce this issue?
 - A. Dynamic memory allocation
 - B. Priority-based interrupt handling
 - C. Round-robin scheduling
 - D. Task delay functions
- 8 When two RTOS tasks share a common resource, which mechanism prevents race conditions?
 - A. Time slicing
 - B. Semaphore or mutex
 - C. Context switching
 - D. Task preemption
- 9 In an embedded system requiring reliable communication with error detection in an automotive environment, which protocol is most suitable?
 - A. I²C
 - B. SPI
 - C. CAN Bus
 - D. UART
- 10 During communication analysis of a network-based embedded design, which parameter directly affects system performance?
 - A. Program memory size
 - B. Network latency
 - C. Number of I/O ports
 - D. Instruction set architecture
- 11 While designing an embedded printer controller, which memory type is most suitable for storing firmware that may require updates?
 - A. Mask ROM
 - B. SRAM
 - C. Flash memory
 - D. Cache memory
- 12 CRC in embedded systems is mainly used to:
 - A. Increase memory speed
 - B. Compress data
 - C. Detect data transmission or storage errors
 - D. Allocate memory blocks

Q2. Solve the following.

- A) An alarm clock is to be designed using an ARM processor. Apply suitable embedded system architecture and explain how memory organization and bus configuration influence system performance. 6
- B) Given an embedded application requiring multiple peripherals with different data rates, apply the concept of scalable bus architecture and justify its selection over a single shared bus. 6

- Q3. Solve the following.
- A) Apply the working principle of an ADC interfaced with a microcontroller to measure temperature from an analog sensor. Explain the role of ports and timing considerations. 6
 - B) A robotic system requires bidirectional motor control. Apply the use of an H-Bridge circuit and explain how it controls direction and speed of a DC motor. 6
- Q4. Solve Any Two of the following.
- A) Apply RTOS task scheduling techniques to a system with periodic and aperiodic tasks. Explain how deadlines can be met using priority-based scheduling. 6
 - B) In a real-time embedded system, shared resources cause data inconsistency. Apply RTOS synchronization mechanisms to solve this problem with suitable justification. 6
 - C) Apply the concept of interrupt processing and clocking in an RTOS to explain how real-time response is achieved in time-critical applications. 6
- Q5. Solve Any Two of the following.
- A) Apply the CAN bus protocol for communication in an automotive embedded system and explain how it ensures reliable data transfer. 6
 - B) An elevator controller uses a distributed embedded architecture. Apply communication analysis and scheduling to explain how system performance is improved. 6
 - C) Apply the selection criteria between I²C and Ethernet for an embedded network-based design, considering speed, distance, and application requirements. 6
- Q6. Solve Any Two of the following.
- A) Apply the steps involved in specification and architectural design of a set-top box embedded system, highlighting processor and memory selection. 6
 - B) Given an embedded system with multiple external peripherals, apply the concepts of memory map, I/O map, and interrupt map to organize system resources efficiently. 6
 - C) Apply the use of Flash memory and CRC in embedded hardware to ensure reliable firmware storage and error detection. 6

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